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Before this lab I thought a more complex model would always fit better. The plots showed that’s not true. The degree 1 line was too simple and missed the sine wave, and the degree 15 line got really wiggly and followed the noisy points. Degree 4 followed the wave without bending to every point, so it was the best fit. The learning curves helped me most. For degree 1, the training and cross-validation scores were both low and close together, which means high bias. For degree 15, the training score was high but the cross-validation score was much lower, which means high variance. This lab helped me see that a model needs to be complex enough to catch the pattern but not so complex that it memorizes the noise.